# Lab 1 – Product Description

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## 1 Introduction

Workforce expectations are rapidly evolving, driven by rapid technological advancement and the increasing integration of AI across industries. As a result, employers are facing significant challenges in identifying candidates with the right skills, with 63% citing skill gaps as a top barrier to growth (World Economic Forum, 2025). At the same time, organizations are adjusting their hiring strategies—66% of enterprises report reducing entry-level hiring due to AI, while 91% indicate that automation is actively changing or eliminating jobs (Laurent, 2026). These shifts are contributing to widespread employment challenges for recent graduates, with an estimated 50–60% experiencing underemployment and 48% feeling underqualified for entry-level roles (Cengage Group, 2025). As illustrated in Figure 1, perceptions of preparedness among recent graduates continue to fluctuate, reflecting uncertainty in how well higher education aligns with workforce demands.

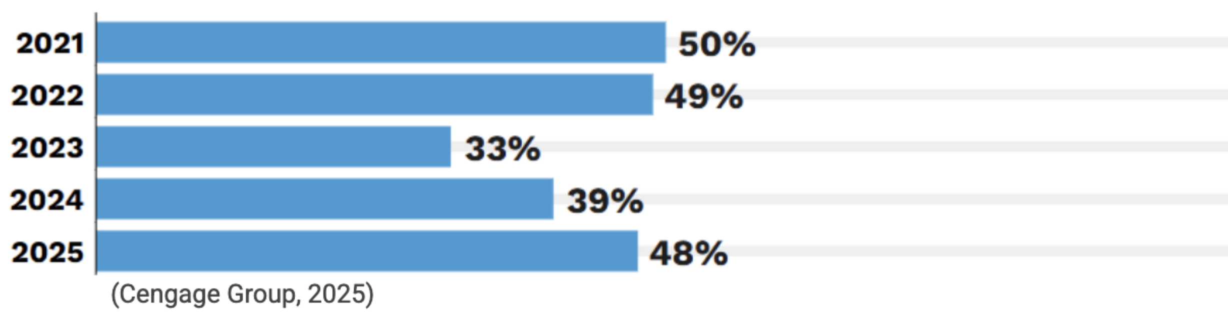


**Figure 1: Fluctuations in Perception of Preparedness in Recent Graduates**

This problem is not isolated to the job market alone but extends across higher education institutions and career services. While academic progress is typically well-tracked, career preparation remains fragmented. Existing tools tend to address isolated needs—such as coursework tracking systems that focus only on academic data, job boards that list opportunities without personalized guidance, and networking platforms that emphasize connections without structured development. This disjointed ecosystem creates a critical gap: there is no unified, real-time system to guide students in building and assessing their career readiness.

Several key characteristics define this problem. Career development systems rely on fragmented data, making it difficult to form a complete picture of a student's preparedness. Gap detection is often delayed, occurring only after missed opportunities or unsuccessful applications. As a result, career decision-making becomes reactive rather than proactive, and students have limited visibility into their true readiness for the workforce. From an educator's perspective, as shown in Figure 2, there is also ongoing ambiguity around the role institutions should play in career preparation versus academic instruction.

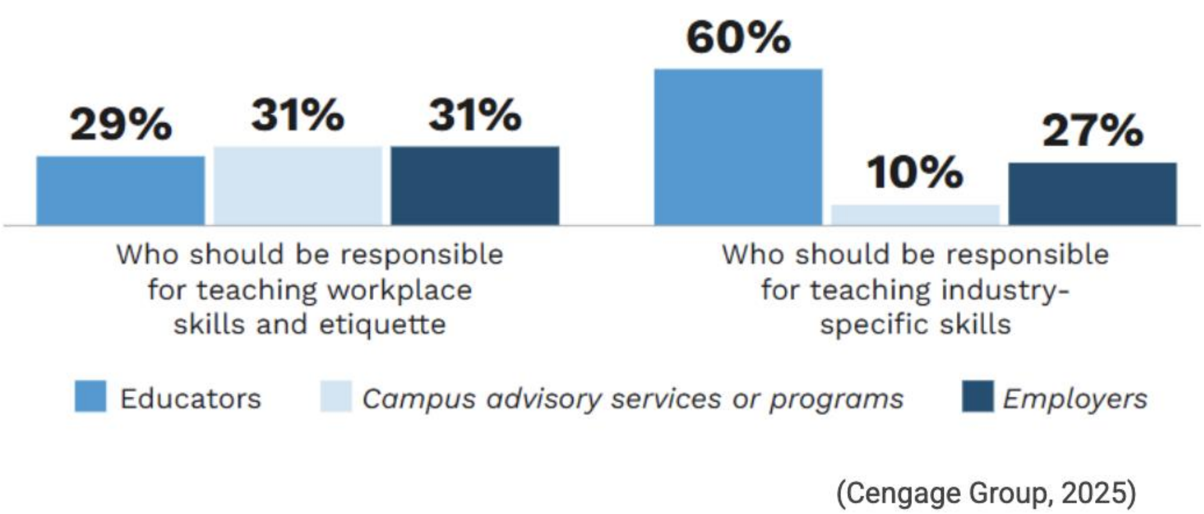


**Figure 2: Educator POV on Teaching Responsibilities**

To address these challenges, GradPace is introduced as a unified, AI-driven career readiness platform designed to bridge the gap between academic progress and professional development, providing students with real-time insights, personalized roadmaps, and actionable guidance toward achieving career success

# 2 Product Description

- Provide a top-level description of CS 410 product for the average reader.
- Provide a summary of the solution –- and its goals and objectives. This section should be one paragraph minimum.

## 2.1 Key Product Features and Capabilities

- What does it do?
- What is significant/unique/innovative about it?
- What does it accomplish?
- Describe how this solves the problem.

## 2.2 Major Components (Hardware/Software)

- Provide an overview of the hardware needed to support the solution.
- Describe how it is structured based on CS 410 MFCD.
- Define and describe the software to be developed.

# 3 Identification of Case Study

- For whom is this product being developed? Why?
- Identify case study group—the small group of users who will use app prototype and provide feedback.
- Who else might use this in the future?

| Features | Prototype | Real World Product |
|---|---|---|
| Login/Authentication | Full: Core functionality and easy to implement | Secure authentication system |
| Account Creation/Deletion | Full: Required for all users | Full account lifecycle management |
| Account Subscription Management | Partial: Simulated or mocked payments | Paid subscription system |
| Input Academic and Professional Info | Full: Core input for system logic | User profile with professional + academic data |
| Personalized Roadmap Generation | Partial: Simplified logic instead of full AI | Career roadmap |
| View Roadmap | Full: UI-based and feasible | Interactive roadmap UI |
| Early-Warning Gap Detection | Partial: Rule-based instead of AI driven | Skill gap analysis |
| Industry Readiness Dashboard | Partial: Basic scoring instead of dynamic benchmarking | Visual readiness scoring system |
| Recommendations/Guided Next Steps | Full: Static or rule-based suggestions | Guided recommendations |
| Notifications/Alerts | Partial: Basic notifications only | Alerts and reminders |
| Location-Based Event Recommendations | Partial: Static or mock event data and smaller AI model for recommendations | GPS + scraped local opportunities |
| Input Up-To-Date Industry Data | Eliminated: Not feasible for prototype | Continuous industry data input system |
| Alumni Survey | Partial: Simple form implementation | Post-graduation feedback system |
| Survey Results Dashboard | Partial: Basic visualization only | Aggregated alumni feedback dashboard |
| User Management | Partial: Limited admin interface | Admin control panel |
| AI Configuration/Rule Updates | Eliminated: Too complex for prototype | Dynamic AI tuning system |
| Data Quality Monitoring | Partial: Manual or basic validation | Automated data validation system |

**Table 1: Table of Comparison Between Real World Product and Prototype**

### 3.1 Prototype Development Challenges

- Describe the expected challenges to be encountered while completing the prototype – e.g., knowledge missing, capability missing, supporting technology issues.

# 4 Glossary

**API (Application Programming Interface):** A set of rules that allows different software applications to communicate and exchange data with each other.

**Authentication:** The process of verifying a user's identity before allowing access to a system.

**Backend:** The server-side part of a website or application that handles logic, databases, and data processing.

**Career Readiness:** A measure of how prepared a student is to enter a specific career based on skills, experience, and qualifications.

**Career Roadmap:** A structured, visual plan that outlines a student's progress, goals, and required steps toward a specific career path.

**Cloud Computing:** The use of remote servers on the internet to store data and run applications instead of a local computer.

**Data Leak:** The accidental or unauthorized exposure of sensitive or private information.

**Database:** An organized collection of data that can be stored, accessed, and managed electronically.

**Encryption:** A security method that converts data into a coded form so that only authorized parties can read it.

**Frontend:** The user-facing part of a website or application that users interact with in their browser.

**Gap Detection (Early-Warning System):** A feature that identifies missing skills, experiences, or qualifications needed for a target career.

**Industry Readiness Benchmarking:** The process of comparing a student's profile against expected skills and qualifications for specific careers.

**Location-Based Event Recommendations:** Suggestions for nearby internships, networking events, or career opportunities based on user location.

**Professional Data Input:** User-provided information such as coursework, projects, internships, and certifications used to build a profile.

**Survey Results Dashboard:** A feature that displays aggregated insights from alumni data to inform students and administrators.

**System Administrator:** A role responsible for managing the platform, maintaining data quality, and overseeing system functionality.

**Token (AI):** A small unit of text processed by an AI model, such as a word, part of a word, or punctuation.

**User Interface (UI):** The visual elements of a software application that allow users to interact with it.

**Web Server:** A computer system or software that stores websites and delivers web pages to users when they request them through a browser.

# 5 References

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